

Neuromaps-PRIME: a graph-based framework for integrating cross-species neuroimaging spaces

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Summary

Brain imaging data are frequently reported in standardized coordinate spaces so that measurements from different individuals can be aligned and compared. The cost of this convention is fragmentation: many different standard spaces are in routine use, and a brain map defined in one space must be transformed before it can be compared with a map defined in another. The problem multiplies once non-human primate (NHP) species are considered alongside human data because of the many NHP coordinate spaces that are in use. Moreover, even within a single coordinate system, the same data may be stored as a cortical surface, a volume, or both, and at varying resolutions. Neuromaps-PRIME is a Python package that treats this as a graph problem: each template space is a node, each transformation an edge, and moving a map from one space to another becomes a matter of finding and applying a path through the graph. The package extends the neuromaps toolbox ([Markello et al., 2022](#)) and currently ships twelve template spaces across 3 primate species — human, macaque, and chimpanzee — together with curated transformations between them. New spaces, transformations, and annotations are declared in configuration files rather than code.

Statement of need

Reorganizing and resampling brain data into a common space is a tedious, error-prone prerequisite for almost any quantitative analysis. The redundant preprocessing it demands slows research and undermines reproducibility. This burden falls hardest on comparative and translational neuroimaging, where interest in NHP models is driven by their close neuroanatomical similarity to humans ([Milham & others, 2018](#)) and where the number of standard coordinate spaces across the macaque, marmoset, and chimpanzee is large and still growing.

Neuromaps-PRIME is built for researchers who need to move brain maps reliably between coordinate spaces or species. It lets a user request any available brain map in any connected space and returns the resampled map, automatically assembling the intermediate steps through the graph when no direct transformation exists. Because spaces and transformations are described in editable configuration files, a group with a custom atlas or unpublished data can register it against the existing network without modifying the software, lowering the barrier to integrating new resources into a reproducible workflow.

State of the field

The most closely related tool is neuromaps (Markello et al., 2022), which popularized programmatic comparison of brain maps but was designed around the human cortex. Its coordinate spaces are hard-coded; it implements only direct pairwise transformations and cannot compose a chain of transforms to bridge spaces that are not directly connected; and it supports surface data only. TemplateFlow (Ciric et al., 2022) addresses an adjacent need — a FAIR archive that makes templates and atlases findable and downloadable across species — but it catalogs resources rather than transforming data between them. Lower-level engines such as the Connectome Workbench (Marcus et al., 2011) and ANTs (Avants et al., 2011) perform individual resampling and registration operations, and cross-species correspondences exist as standalone datasets (Sirmipilatz et al., 2020; Xu et al., 2020), but none of these provides an integrated layer that selects, composes, and applies the right sequence of transforms on demand. RheMAP (Sirmipilatz et al., 2020) and RheMAP-Surf (Zhou & Xu, 2025) are data repositories that provide standard transformations, but only within macaque spaces.

Rather than re-implement this ecosystem, Neuromaps-PRIME extends the neuromaps model and orchestrates these established tools and datasets. Its distinct scholarly contribution is the graph engine itself together with first-class support for multiple species and for volumetric as well as surface data — capabilities that the hard-coded, pairwise, surface-only design of neuromaps cannot accommodate without a fundamental redesign.

Software design

The package centers on a NeuromapsGraph, a directed multigraph (built on networkx (Hagberg et al., 2008)) whose nodes are template spaces, each carrying its species, surfaces, volumes, and annotations, and whose edges are surface or volume transformations. The graph approach was necessary because enumerating every pairwise transform—as neuromaps does—grows quadratically with the number of spaces and quickly becomes unmaintainable. By contrast, adding a node to a graph requires only a single edge to an existing space, after which any other space becomes reachable by path-finding. Given a source and target, the engine finds the shortest weighted path, composes the intermediate transforms, and can insert the composed result back into the graph as a cached edge, trading a small amount of storage for faster repeated queries. Four transformers cover the common operations — surface-to-surface and volume-to-volume resampling, and surface-to-volume and volume-to-surface projection — for both continuous (metric) and discrete (label) data.

Two further design decisions support reproducibility and community growth. First, the network is described in editable YAML files validated against bundled JSON schemas, separating configuration from implementation so that contributors can add spaces and transforms declaratively; data themselves are hosted remotely and fetched on demand into a local cache, avoiding the need to ship large template files. Second, because the underlying transformations require specific versions of external neuroimaging tools, these are invoked through NiWrap (Rupprecht et al., 2025), which can dispatch to local, Docker, Podman, or Singularity/Apptainer runtimes; this isolates the library from environment-specific version fragility at the cost of an optional container dependency. The transformations themselves draw on Multimodal Surface Matching (Robinson et al., 2014), the CIVET environment (Lepage et al., 2017), RheMAP volumetric warps (Sirmipilatz et al., 2020), and cross-species functional alignment between the human and macaque cortex (Xu et al., 2020).

Research impact statement

Neuromaps-PRIME (v0.1.0) is released under the MIT license, distributed on PyPI, documented online, and maintained with a continuous-integration suite spanning unit, integration, and

regression tests. It builds directly on neuromaps, an established and widely cited tool in the human neuroimaging community, and extends that ecosystem to the NHP imaging community organized around resources such as PRIME-DE (Milham & others, 2018). At release the graph integrates twelve template spaces across three species with curated surface and volume transformations, and its configuration-driven design has been exercised through community development activities aimed at adding further spaces and brain maps. We anticipate the framework will serve as reusable infrastructure for cross-species comparative analyses; concrete adoption metrics and dependent publications will accrue as the software matures.

AI usage disclosure

Generative AI tools were used to assist in drafting and copy-editing this manuscript. All AI-assisted text was reviewed and verified for accuracy by the authors against the software's source code, documentation, and the cited literature, and the authors take full responsibility for its content.

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References

- Avants, B. B., Tustison, N. J., Song, G., Cook, P. A., Klein, A., & Gee, J. C. (2011). A reproducible evaluation of ANTs similarity metric performance in brain image registration. *NeuroImage*, 54(3), 2033–2044. <https://doi.org/10.1016/j.neuroimage.2010.09.025>
- Ciric, R., Thompson, W. H., Lorenz, R., Goncalves, M., MacNicol, E. E., Markiewicz, C. J., Halchenko, Y. O., Ghosh, S. S., Gorgolewski, K. J., Poldrack, R. A., & Esteban, O. (2022). TemplateFlow: FAIR-sharing of multi-scale, multi-species brain models. *Nature Methods*, 19(12), 1568–1571. <https://doi.org/10.1038/s41592-022-01681-2>
- Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). *Exploring network structure, dynamics, and function using NetworkX*. 11–15.
- Lepage, C., Lewis, L., Jeun, S., Bermudez, P., Khalili-Mahani, N., Omidyeganeh, M., Zijdenbos, A., Vincent, R. D., Adalat, R., & Evans, A. C. (2017). Human MR evaluation of cortical thickness using CIVET v2.1. *Organization for Human Brain Mapping Annual Meeting*.
- Marcus, D. S., Harwell, J., Olsen, T., Hodge, M., Glasser, M. F., Prior, F., Jenkinson, M., Laumann, T., Curtiss, S. W., & Van Essen, D. C. (2011). Informatics and data mining tools and strategies for the Human Connectome Project. *Frontiers in Neuroinformatics*, 5, 4. <https://doi.org/10.3389/fninf.2011.00004>
- Markello, R. D., Hansen, J. Y., Liu, Z.-Q., Bazinet, V., Shafiei, G., Suárez, L. E., Farahani, N., & Misić, B. (2022). Neuromaps: Structural and functional interpretation of brain maps. *Nature Methods*, 19(11), 1472–1479. <https://doi.org/10.1038/s41592-022-01625-w>
- Milham, M. P., & others. (2018). An open resource for non-human primate imaging. *Neuron*, 100(1), 61–74.e2. <https://doi.org/10.1016/j.neuron.2018.08.039>
- Robinson, E. C., Jbabdi, S., Glasser, M. F., Andersson, J., Burgess, G. C., Harms, M. P., Smith, S. M., Van Essen, D. C., & Jenkinson, M. (2014). MSM: A new flexible framework for multimodal surface matching. *NeuroImage*, 100, 414–426. <https://doi.org/10.1016/j.neuroimage.2014.07.061>

- 131 [neuroimage.2014.05.069](#)
- 132 Rupprecht, F. J., Kai, J., Shrestha, B., Giavasis, S., Xu, T., Glatard, T., Milham, M. P., &
133 Kiar, G. (2025). Styx: A multi-language API generator for command-line tools. *bioRxiv*.
134 <https://doi.org/10.1101/2025.07.24.666435>
- 135 Sirmpilatze, N., Klink, P. C., & Messinger, A. (2020). *RheMAP: Non-linear warps between*
136 *common rhesus macaque brain templates*. Zenodo. [https://doi.org/10.5281/zenodo.](https://doi.org/10.5281/zenodo.4748589)
137 [4748589](https://doi.org/10.5281/zenodo.4748589)
- 138 Xu, T., Nenning, K.-H., Schwartz, E., Hong, S.-J., Vogelstein, J. T., Goulas, A., Fair, D. A.,
139 Schroeder, C. E., Margulies, D. S., Smallwood, J., Milham, M. P., & Langs, G. (2020).
140 Cross-species functional alignment reveals evolutionary hierarchy within the connectome.
141 *NeuroImage*, 223, 117346. <https://doi.org/10.1016/j.neuroimage.2020.117346>
- 142 Zhou, Z.-X., & Xu, T. (2025). *RheMAP-Surf: Surface transformations among commonly used*
143 *rhesus macaque brain templates*. Zenodo. <https://doi.org/10.5281/zenodo.15642802>

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